



Parcel Tools

User Guide

Professional land-surveying software: area calculation, point files and CAD import

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1. Welcome to Parcel Tools

Parcel Tools helps land surveyors turn a points file (the coordinates measured in the field) and a CAD drawing into **parcel areas, perimeters and ready-to-print PDF reports** - quickly and without calculation mistakes.

With Parcel Tools you can:

- Load a points file (.pnt, .txt or .csv) and edit it.
- Build a parcel by choosing its corner points and get the **area and perimeter** instantly.
- Add **curves** to a parcel using the middle-ordinate (M) method.
- Open a **DWG or DXF** drawing, click a parcel (polygon or hatch) and let the program find its corners and match them to your points.
- View all your points and parcels on a **plot**.
- Export one parcel, or many parcels together, to a **PDF report**.

- Distribute the area error against the registered title area (**Error Calculations**).

Note

Everything is saved on your own computer. Your drawings and points are not uploaded anywhere.

This guide walks you through the program step by step. If you are in a hurry, read **Chapter 4 (Quick start)** first - it takes five minutes.

2. Installation and first start

System requirements: Windows 10 or Windows 11 (64-bit). An internet connection is needed for signing in, license activation and automatic updates. To open **DWG** files you also need **AutoCAD** or the free **ODA File Converter** installed on the computer (DXF files need nothing extra).

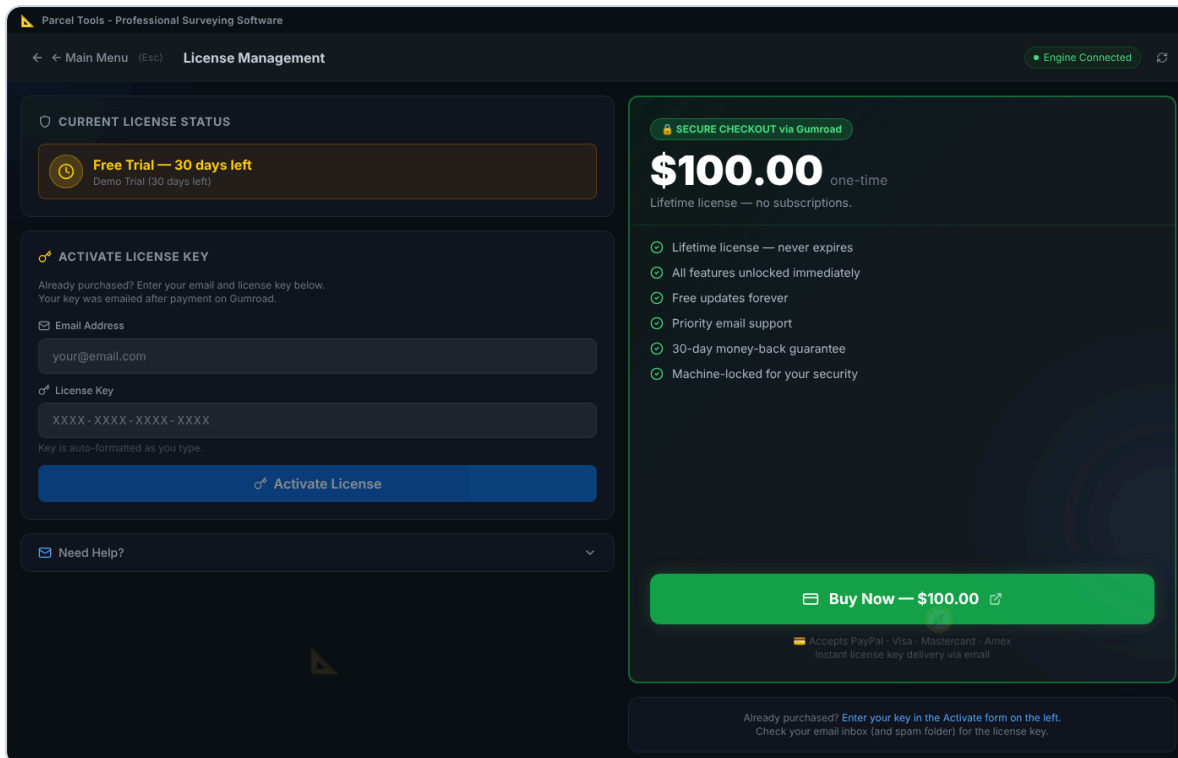
- 1 Open the Parcel Tools website and click **Download for Windows**. The file is named **Parcel-Tools-Setup-2.0.20.exe**.
- 2 Double-click the downloaded file. If Windows shows a blue "Windows protected your PC" window, click **More info** and then **Run anyway**.
- 3 Follow the installer, then open **Parcel Tools** from the Start menu or the desktop icon.
- 4 On first start, create an account (next chapter).

Tip

Updates are automatic. When a new version is published, Parcel Tools tells you inside the program. Click the update message, wait for the download, and choose **Restart** - your projects are not touched.

3. Account and license

When you open the program for the first time you are asked to **sign up** (email and password) or **log in**. A new account starts with a **free 30-day trial** with all features unlocked. The days left are shown at the top of the main menu and in the status bar.



The License page: current status, key activation and purchase.

To buy and activate a license:

- 1 Open **License** from the main menu (Quick Actions).
- 2 Click **Buy Now** and complete the secure checkout (Gumroad). A one-time payment gives a lifetime license with free updates.
- 3 Your **license key** arrives by email (check your spam folder too).
- 4 Back in the program, type your email and the key in **Activate License** and click the button. The key is formatted automatically as you type.

Note

The license is tied to one computer for your security. If you change computers, contact support (see the last page).

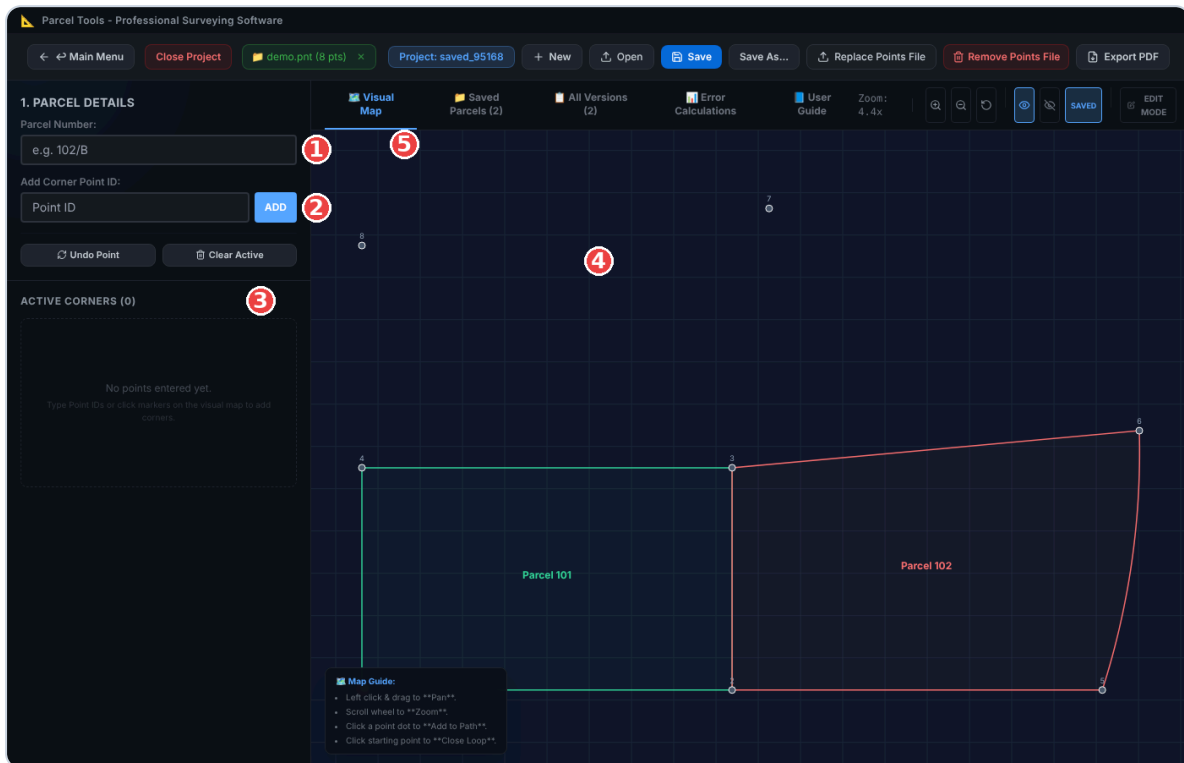
4. Quick start - your first parcel in 5 steps

This is the fastest way to a result. Every step is explained in detail in the next chapters.

- 1 **Main menu → Area Calculator.**
- 2 Click **Load Points** and choose your points file (each line: ID, X, Y).
- 3 Type a **parcel number**, then type the corner point numbers one by one, pressing **Enter** after each one (or click the points on the map). To close the parcel, enter the **first point**

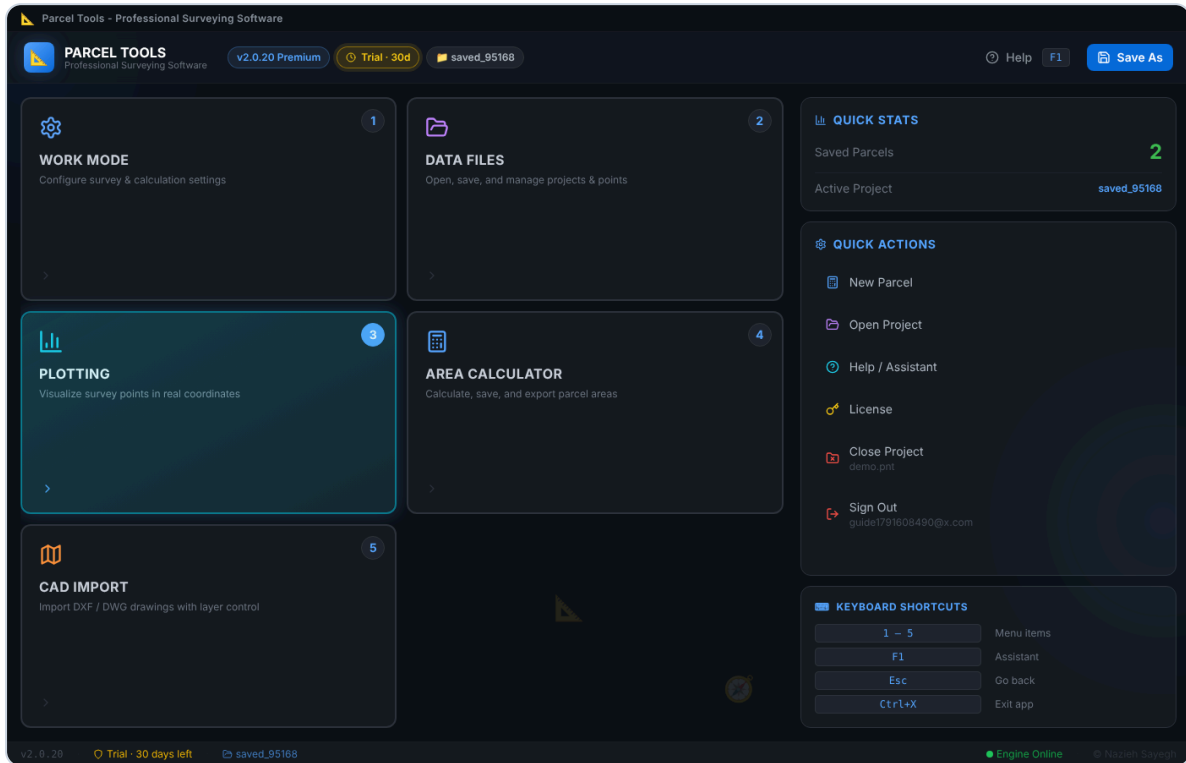
again.

- 4 Answer the question about curves: **Cancel** if there are none. The **area and perimeter** appear.
- 5 Click **Save Parcel**. Use **Export PDF** to get the report.



The Area Calculator after two parcels were saved. (1) Parcel number, (2) point ID box, (3) list of active corners, (4) visual map, (5) tabs: saved parcels, versions, error calculations and help.

5. The main menu



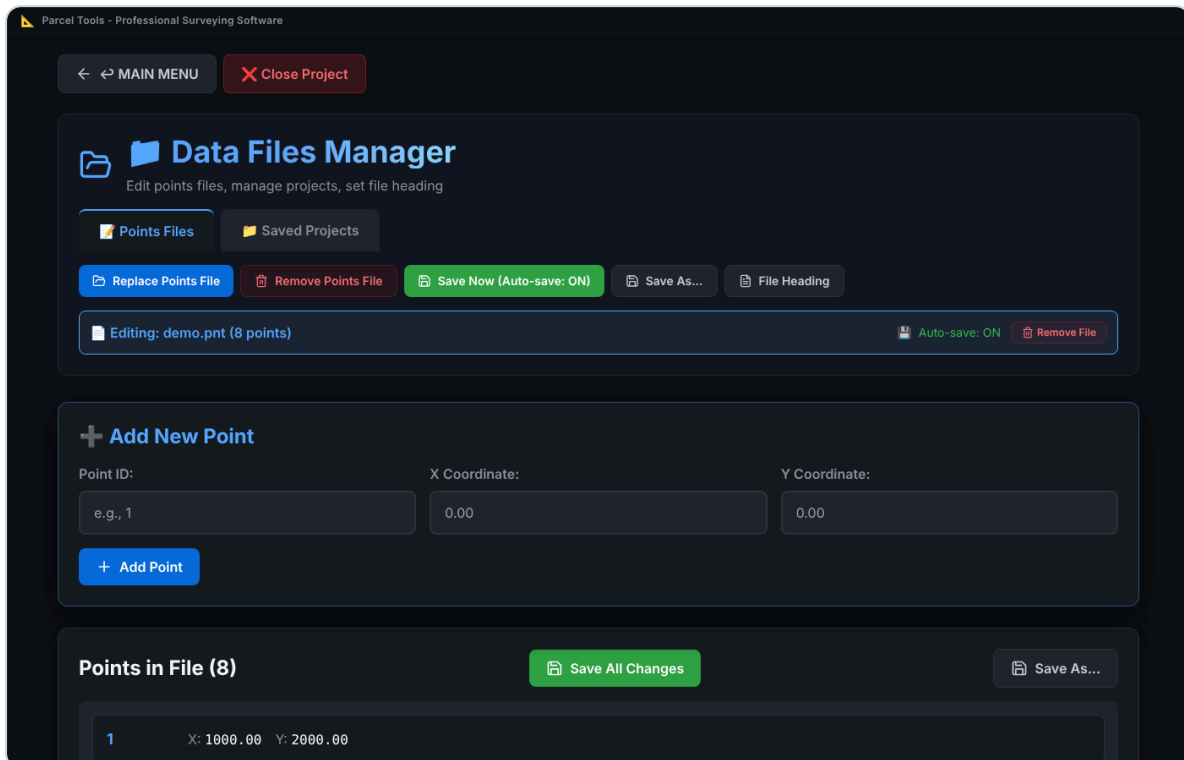
The main menu.

Item	What it is for
1 - Work Mode	Coordinate system, units, decimal precision and the assistant settings.
2 - Data Files	Open and edit points files, manage saved projects, set the file heading used in PDFs.
3 - Plotting	Draw all points and parcels to scale.
4 - Area Calculator	Build parcels, add curves, save and export.
5 - CAD Import	Open DWG / DXF drawings, pick parcels and create them with matched points.

On the right: **Quick Stats** (saved parcels and the active project) and **Quick Actions** (New Parcel, Open Project, Help, License, Close Project, Sign Out).

Shortcut	Action
1 to 5	Open that menu item
F1	Open the Assistant (help)
Esc	Go back / return to the main menu
Ctrl + X	Exit the program

6. Data Files - your points



Data Files Manager: edit the points of the loaded file.

A **points file** is a text file with one point per line:

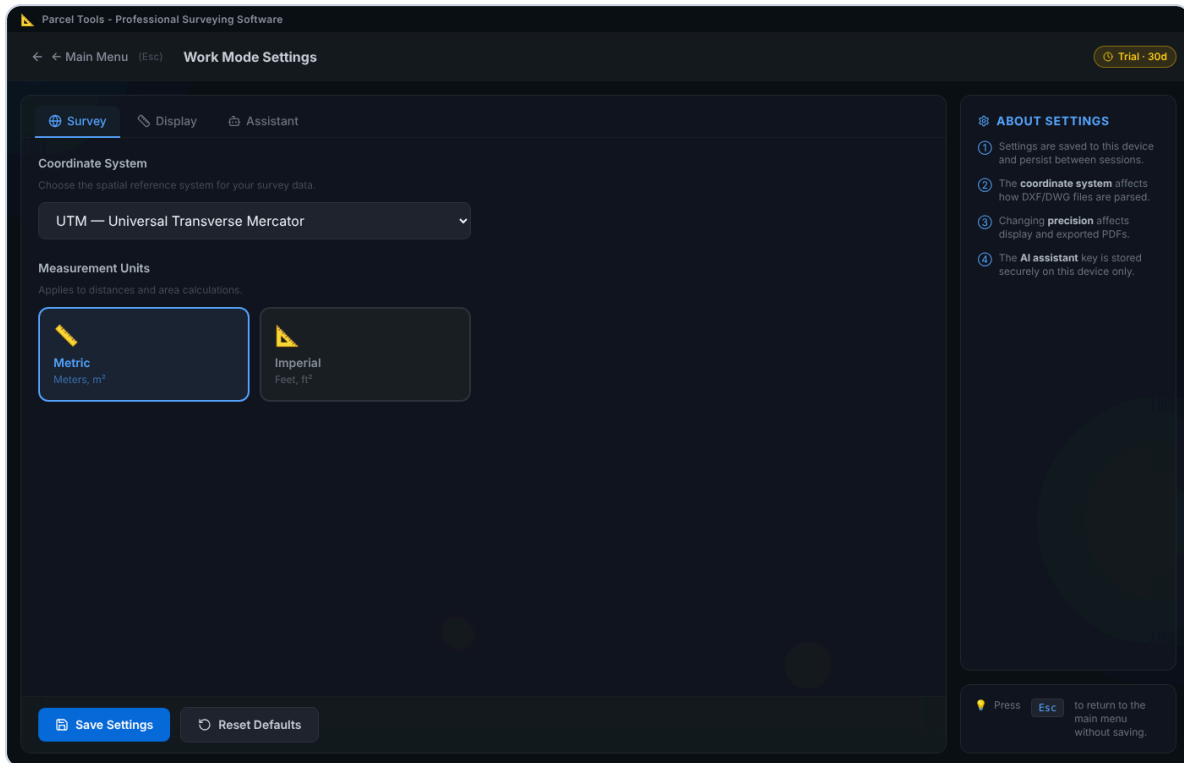
1, 1000.00, 2000.00 - point number, X coordinate, Y coordinate. Commas or spaces are both accepted. Supported files: **.pnt**, **.txt**, **.csv**.

- **Open Points File** loads a file. **Replace / Remove Points File** changes or unloads it.
- **Add New Point** adds a point; every point in the list has buttons to edit or delete it.
- **Save Now / Save All Changes** writes back to your file and keeps its format. **Auto-save** is ON by default.
- **File Heading** stores block, quarter and parcel information that can be printed in the PDF.
- The file is watched: if you change it in another program the changes are picked up.
- The **Saved Projects** tab lists your project files (.prcl) so you can reopen them.

Tip

Use unique point numbers. If the same number appears twice the program warns you, because the area could be wrong.

7. Work Mode - settings



Work Mode settings.

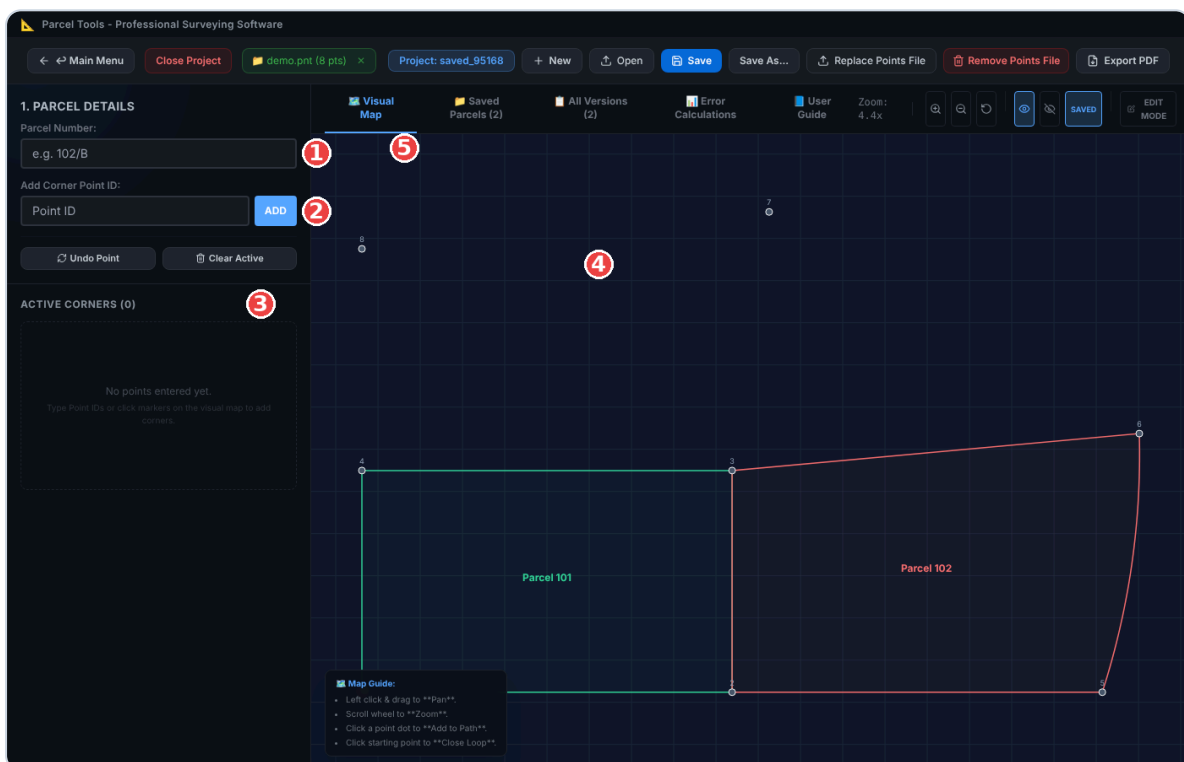
- **Survey tab:** choose the coordinate system (UTM, Geographic, Local Grid, State Plane) and the units (**Metric** m / m² or **Imperial** ft / ft²).
- **Display tab:** decimal precision for numbers on screen and in PDFs.
- **Assistant tab:** optional settings for the built-in assistant. The key is stored only on your computer.

Click **Save Settings** to keep your choices, or **Reset Defaults** to go back. Settings are remembered between sessions. Press **Esc** to leave without saving.

8. Area Calculator in detail

8.1 Building a parcel

- 1 Load your points (**Load Points** at the top, or open a project).
- 2 Enter the **Parcel Number** (for example 101 or 102/B).
- 3 Add corners **in order around the parcel**: type the point number and press **Enter** (or click **ADD**). You can also click the points on the **Visual Map**.
- 4 Close the loop by entering the **first point again** (or clicking it on the map).

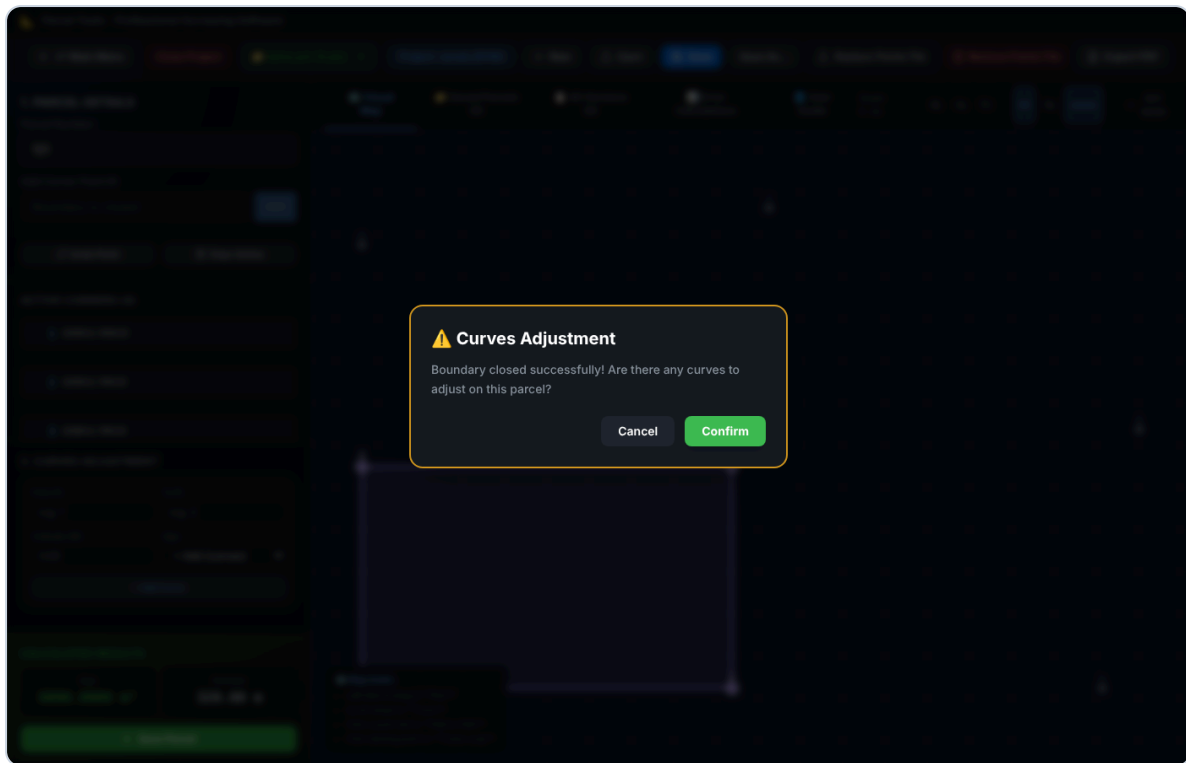


Numbers 1-5 mark: the parcel number, the point box, the corner list, the map and the tabs.

- **Undo Point** removes the last corner. **Clear Active** starts over.
- **Edit**, **Delete** and **+ Insert between** next to each corner let you correct the list without starting again.
- The map: drag to move, scroll to zoom, the round buttons fit the view and hide/show labels.

8.2 Curves (middle-ordinate method)

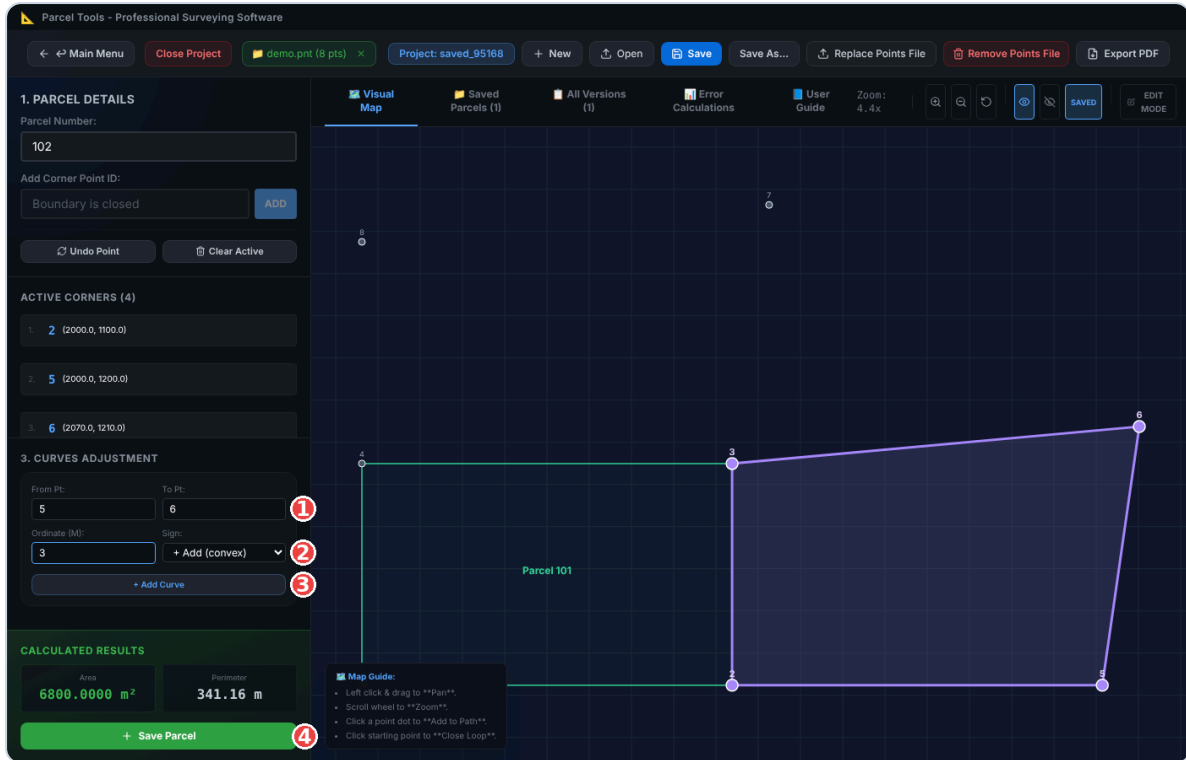
When the boundary closes, the program asks "**Are there any curves to adjust on this parcel?**"



Curves Adjustment question. Choose Cancel if the parcel has no curves.

Choose **Confirm** to add curves:

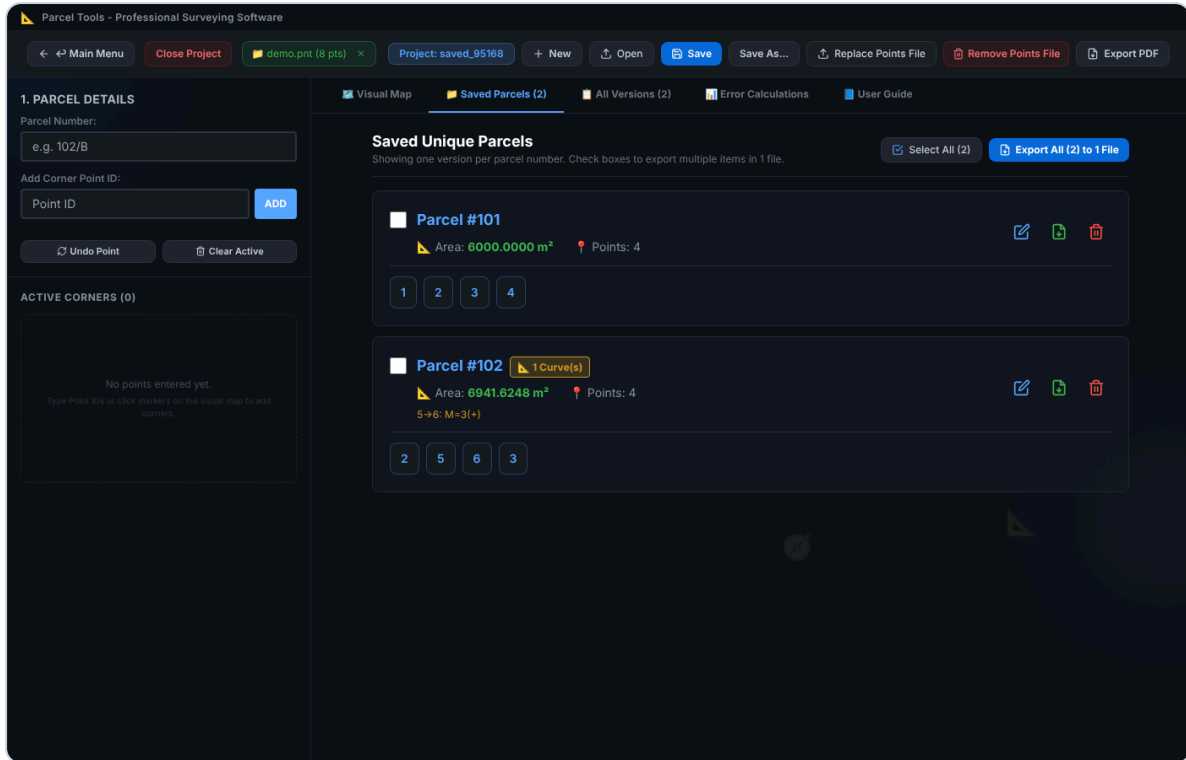
- 1 In **From Pt** and **To Pt**, enter the two corners the curve connects (for example 5 and 6).
- 2 In **Ordinate (M)**, enter the middle ordinate in meters - the distance from the middle of the chord to the arc.
- 3 Choose the **Sign: + Add (convex)** adds the circular segment to the area, - **Subtract (concave)** removes it.
- 4 Click **+ Add Curve**. You can add several curves.



Curve form: (1) from / to points, (2) ordinate and sign, (3) Add Curve, (4) Save Parcel.

8.3 Saving, versions and the tabs

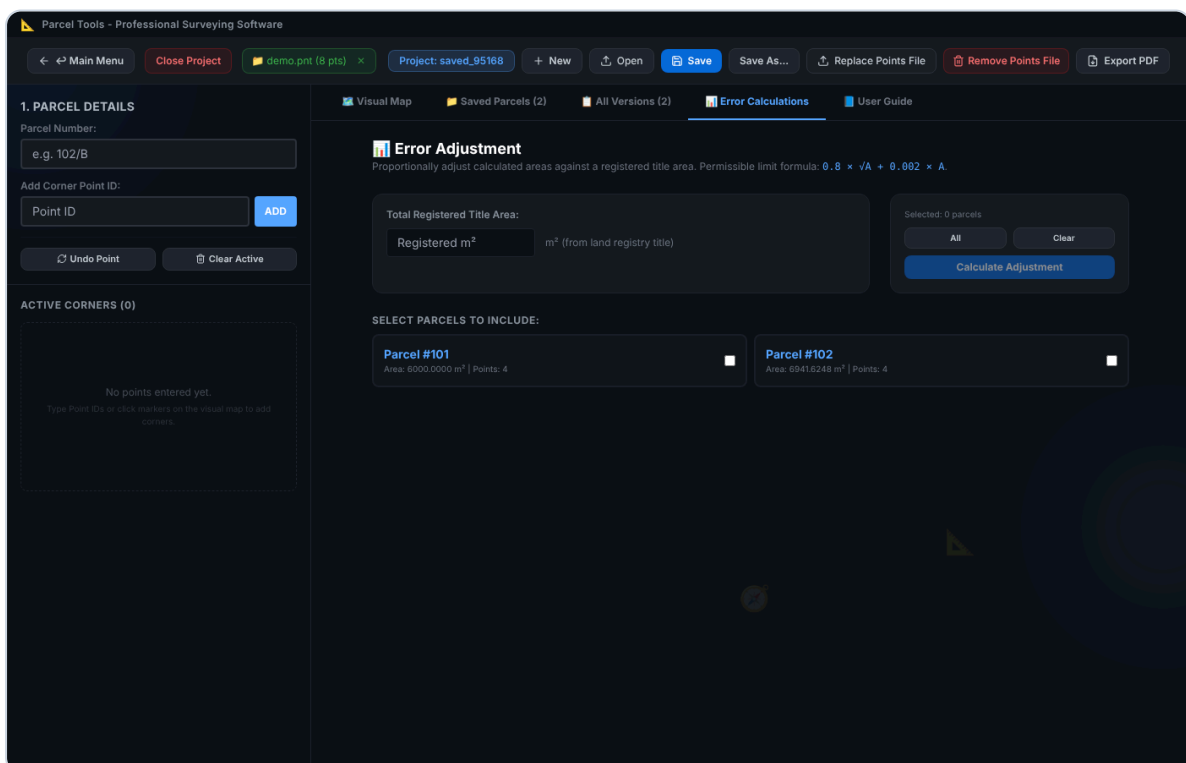
- **Save Parcel** stores the parcel in the project. The project also **auto-saves** every few seconds.
- **Saved Parcels** shows the latest version of every parcel number with its area, points and curves. From each card you can load it back into the editor, export it to PDF, or delete it.
- **All Versions** keeps every save. Saving the same parcel number again creates a new version, so you never lose an earlier result.
- **Visual Map** shows all saved parcels in different colours.



Saved Parcels tab. Tick several boxes and use Export to put them in one PDF.

8.4 Error Calculations

Use this tab to compare your calculated areas with the **registered title area** and spread the difference proportionally. Enter the total registered area, tick the parcels that belong together and click **Calculate Adjustment**. The permissible limit is shown with the formula **$0.8 \times \sqrt{A} + 0.002 \times A$** .

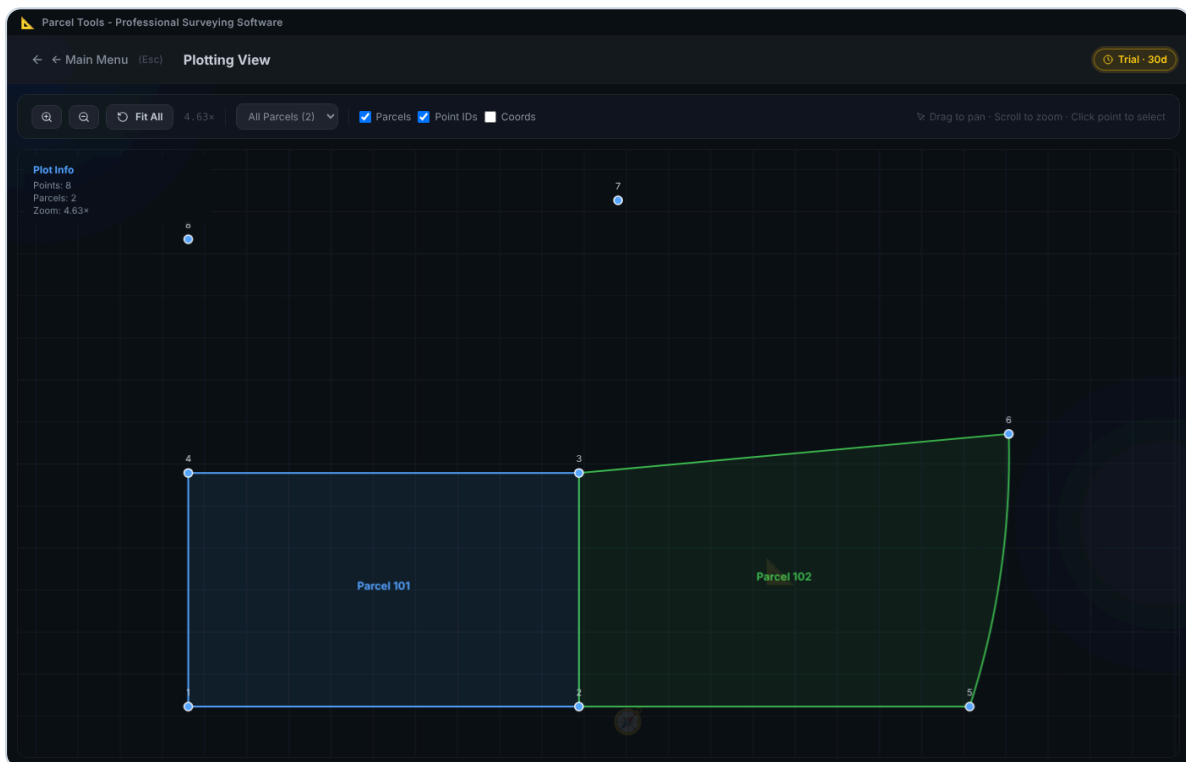


Error Adjustment.

8.5 PDF reports

- **Export PDF** (top bar) exports the saved parcels.
- The green PDF icon on a parcel card exports that single parcel.
- Tick several parcels and click **Export All to 1 File** to get one combined report.
- Tick **Include File Heading on Report Header** to print the block / quarter information from Data Files.

9. Plotting



Plotting view with two parcels.

Plotting draws every point and saved parcel to scale. Use **Fit All** to see everything, the mouse wheel to zoom and drag to move. The checkboxes switch **Parcels**, **Point IDs** and **Coords** on or off, and the list lets you show one parcel only. Click a point to select it.

Note

Plotting needs a loaded points file. If you see "No Points Loaded", open Data Files and load one first.

10. CAD Import (DWG / DXF)

CAD Import reads a drawing, finds the parcel you click, finds its corners, matches them to the numbers in your points file and calculates the area - including curved boundaries and hatches.

Important

A **project with a points file must be open first** (Area Calculator → Load Points, or open a saved project). The CAD page asks for it if none is open.

10.1 Open a drawing and pick a parcel

- 1 Click **OPEN DWG** and choose a `.dwg` or `.dxf` file. A DWG is converted to DXF in the background (this needs AutoCAD or the ODA File Converter - see Chapter 2). If conversion fails, in AutoCAD use **SAVEAS** and save the drawing as **DXF**, then open that file.
- 2 The drawing appears. Scroll to zoom, drag to move.
- 3 Click the parcel (its boundary line or its hatch). It is highlighted and the **LIVE AREA** shows at the top.
- 4 Click **CREATE PARCEL**.

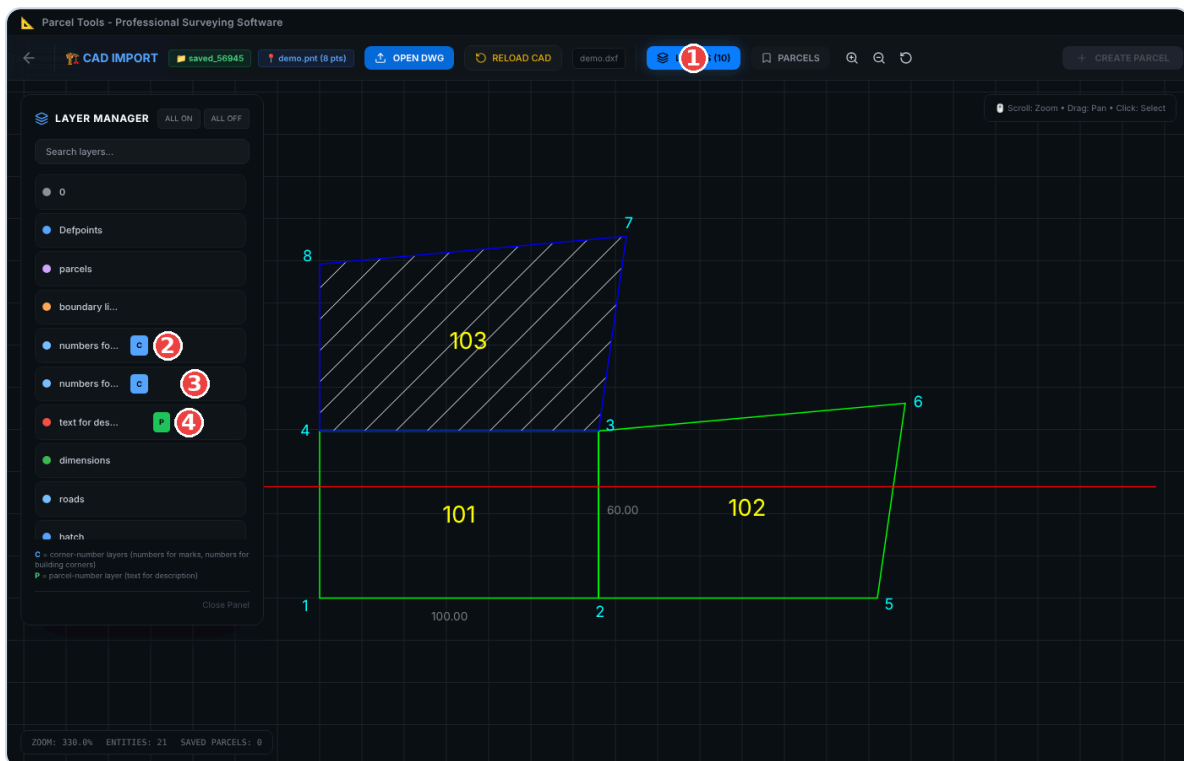


(1) Open drawing, (2) Layers, (3) Create parcel. The hatched area is parcel 103.

10.2 The number layers (very important)

To avoid mistakes, Parcel Tools reads **numbers only from the right layers**. Dimensions, road names and other text on other layers are **ignored**:

Used for	Layer names (detected automatically)
Corner numbers	numbers for marks and numbers for building corners
Parcel number	text for description

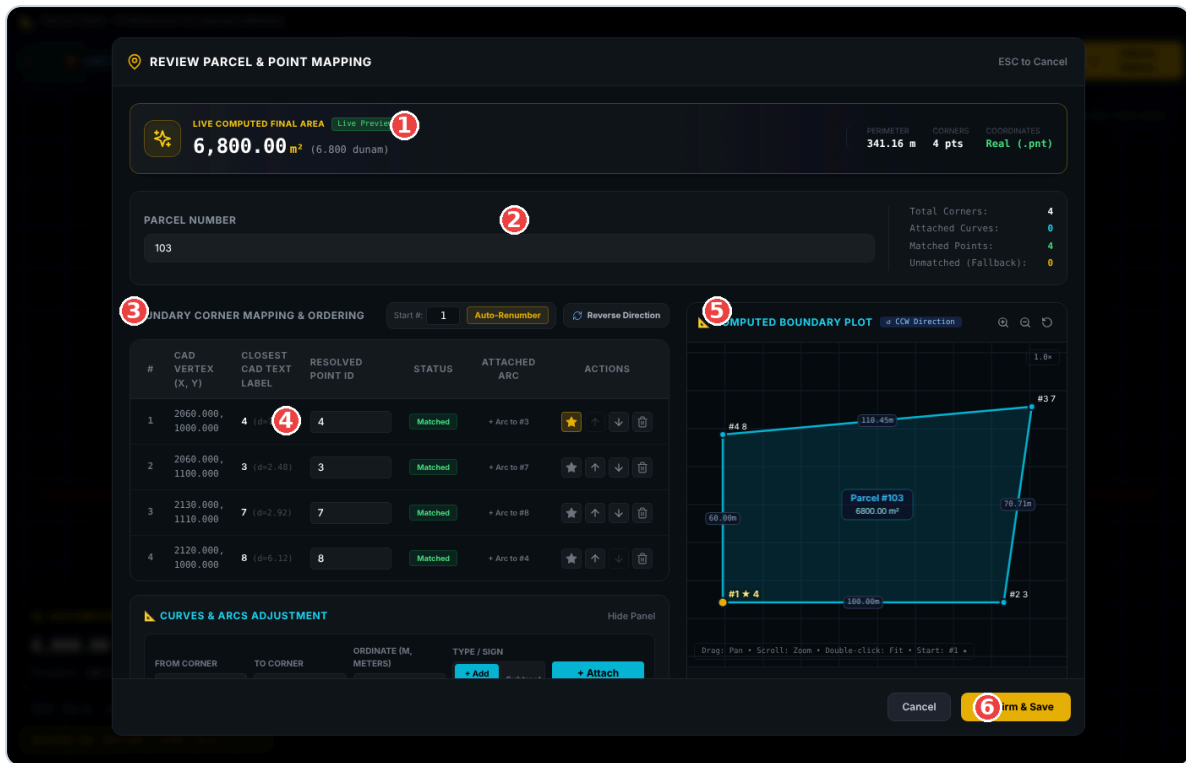


Layer Manager. (1) Open it, (2) blue **C** = corner-number layer, (3) green **P** = parcel-number layer.

- Open **LAYERS** to see the **C** and **P** badges. If your drawing uses different names, click **C** or **P** on the right layers yourself - the program remembers your choice.
- Use **ON / OFF** and **ISO** to hide layers or show one layer only, and the search box to find a layer.
- If no such layers exist in a drawing and none are chosen, the program falls back to reading all text.

10.3 The review window

Before anything is saved you see the **Review Parcel & Point Mapping** window:



(1) Live area, (2) parcel number, (3) corner table, (4) boundary plot, (5) curves panel, (6) Confirm & Save.

- **Live area**, perimeter and number of corners are shown at the top.
- **Parcel number** is taken from the description layer - you can edit it.
- The **corner table** shows each CAD vertex, the closest label, the **resolved point ID** and a **Matched** status. You can change an ID, move a corner up or down, delete a corner, choose the **start corner** (star) or **Reverse Direction**.
- The **plot** on the right shows the shape with the length of every side. Scroll to zoom, drag to move, double-click to fit.
- **Curves & Arcs Adjustment** lets you attach a curve between two corners if one is missing.
- A yellow **warning** appears when a label looks wrong - read it and check that corner.

When everything looks right click **Confirm & Save**. The parcel is added to your project and the Area Calculator.

10.4 Hatches

- Hatches are drawn like in AutoCAD (pattern lines, not a solid colour).
- Curved edges are kept as **true arcs**, so the area is exact.
- A hatch with a **hole** gives the area **minus the hole**.
- If a corner has no point in your file, the program shows it as a new corner so you can decide.

Tip

Always compare the area shown in the review window with the area in AutoCAD for the first parcels of a new drawing. They should match.

11. Projects, saving and sharing

- A **project** (.prcl file) holds your points file reference, parcels, curves and settings.
- Use **Save As** to choose where to keep it. After that, the project **auto-saves**.
- **Open / Open Project** and the **recent files** list on the home screen reopen your work.
- To work on another computer, copy the .prcl file **and** your points file and drawing together.
- Use **Close Project** before opening a different one.

Note

Tip: keep the project, the points file and the drawing of one job in the same folder.

12. Problems and solutions

Problem	What to do
The CAD IMPORT buttons are disabled	Open or create a project with a points file first.
A corner shows "None found"	No label was found next to that corner on the corner-number layers. Check the C layers, or type the point ID in the table yourself.
Warning: "label was ignored"	The label belongs to another parcel. The corner was matched by its position in your points file - check it.
The parcel number is wrong	Check that the P layer is the description layer, or edit the number in the review window.
The area differs from AutoCAD	Check the corner list, the curves and holes. Look for a corner matched to the wrong point.
"File watching unavailable"	The project was not saved yet. Use Save As to choose a place.
The program cannot connect	Wait a few seconds: the calculation engine starts with the program. The status bar shows Engine Online when ready.
A DWG file does not open	Install AutoCAD or the free ODA File Converter, or save the drawing as DXF from AutoCAD (SAVEAS) and open the DXF.
License problem	Open License , check the email and key, or contact support.

13. Updates and support

- The program checks for updates itself and asks before installing.
- Press **F1** for the built-in assistant: ask it how to do anything in the program.
- Support email: **nsayegh2003@gmail.com**. Please send your version number (bottom bar), a short description and, if possible, a screenshot.

Thank you for using Parcel Tools.